Recommended Assessment

Self-Localization

1. Suppose the QBot takes LiDAR scans at each of the locations, A and B, shown below. Describe how the data points in scan B to are related to the points in scan A.

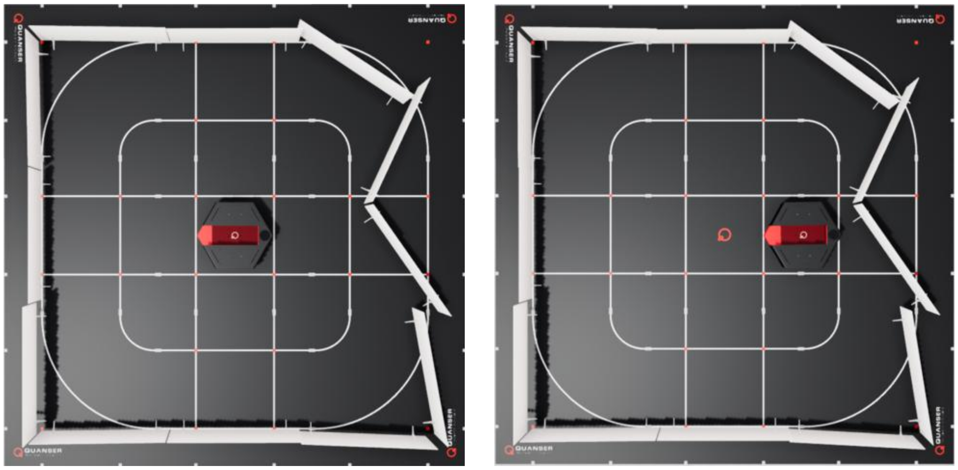

Figure 1. Top view of QBot in location A (left) and location B (right)

2. Suppose the QBot takes LiDAR scans in each of the orientations, C and D, shown below. Describe how you would relate the data points in the scan taken in orientation C to the scan taken in orientation D.

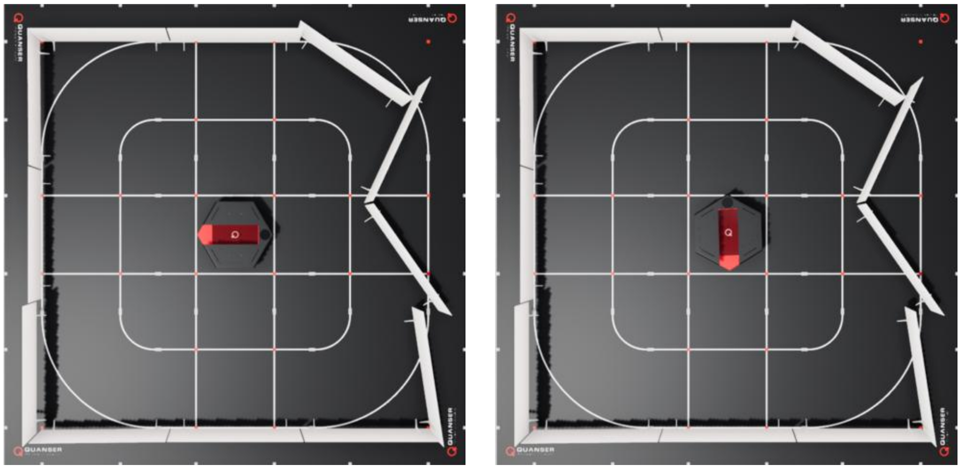

Figure 2. Top view of QBot in orientation C (left) and orientation D (right)

3. Provide a reason for the variation in the main control loop timing shown on the scope display.

4. In this lab, the walls in the environment were modified from previous labs. Discuss how scan matching performance may change if the walls instead formed a square shape.

5. How does the resolution impact performance? Discuss the trade-off in increasing the resolution of the scan match.

6. What did you notice about performance when you decreased the search range to 1 m? What is the trade-off between using a larger or smaller search range? Discuss how you can optimize the search range.

7. What did you notice about the score and the accuracy of the scan match estimate? In this lab, you were able to see when the position estimate was inaccurate by monitoring the plotted XY position on a map. In real-life applications, this is not a practical way to validate the estimate. In that case, why would the score be useful?

8. (For students that completed this lab on both virtual and hardware setups) How did the reference LiDAR scan compare between the virtual and hardware setups? How did this impact the scan match performance? What considerations need to be made about the environment when using LiDAR-based localization?